Brian Chu

Chemical Engineer · Software Engineer

(415)-969-0765 | brian.chu1030@gmail.com | bchu-ops.github.io/pumpkio9-website/ | bchu-ops | brian-chu123

Summary

M.S. Chemical Engineering at UC San Diego specializing in thermal and electrochemical processes, process optimization, and sustainable energy systems. Experienced in lab scale coin cell batteries, materials testing, reactor and heat exchanger design, and data-driven process automation. Built multiple projects combining ML with chemical engineering and automotive engineering with physics-based loss functions.

Engineering Experience & Projects

UC San Diego (Summer Battery Camp) *Jun. 2022 - Aug. 2022*

Battery Engineering Technician *San Diego, CA*

- Fabricated and tested **Li+ coin cells** under controlled conditions, validating improved charge–discharge efficiency
- Formulated and **optimized cathode electrode slurries** balancing porosity, leading to a 15% increase in discharge capacity through binder dissolution, active material dispersion, precision casting, and thermal drying to ensure uniform coating
- Conducted 50+ battery performance and degradation tests, improving feedback cycle time by 30%
- Streamlined testing with **DOE principles, reducing test iteration time by 25%** while maintaining safety compliance
- Drove 25% faster test iterations through improved procedures and risk-assessed workflow refinements

UC San Diego *Jan. 2023 - Jun. 2023*

Thermal Process Engineer *San Diego, CA*

- Designed and simulated an **industrial scale dimethylamine reactor** and heat exchanger system, achieving 30% higher energy efficiency and **95% conversion of methanol** using Aspen Plus & COMSOL
- Formulated piping layout via P&IDs and compliance documents aligning with SEMI safety standards
- Authored RFPs and change controls to support **process improvements by 40%** and ensure regulatory compliance

Vulcan Engineering Solutions *Dec. 2024 - Present*

Software Engineer Intern *Irvine, CA*

- Integrated AI-driven automation into engineering workflows, **increasing process throughput by 60%**
- Developed AI-driven defect detection systems ML pipelines (R-CNN, OpenCV) **improving visual analysis accuracy by 45%**
- Collaborated with **cross-functional teams** to engineer modular pipelines, improving operational efficiency by 50%

Speed Demon *Sept 2025*

Project *Taipei, Taiwan*

- Developed a system to analyze lap times from top racers and quantify input improvements, functioning like a race engineer during qualifying laps
- Designed **FSAE-style go-kart chassis in SolidWorks** and integrated Python, PostgreSQL, OpenAI systems to optimize throttle and steering inputs using an **AWS EC2** server
- Built a persistent memory database for real-time control analytics and performance tuning

NLP for Drug-Drug Interaction Modeling *May 2025*

Project *San Diego, CA*

- **Developed a Knowledge Graph Neural Network** with Python and TensorFlow to predict the combined drug effects

Education

University of California, San Diego *Sept. 2024 - Present*

Master of Science in Chemical Engineering

- Focus: Thermal Processes and Nanoengineering

University of California, San Diego *Sept. 2020 - Jun. 2024*

Bachelor of Science in Chemical Engineering and Cognitive Science w/ ML

- Organizations: ACM, AIChE, Pi Alpha Phi, Chem-E-Car

Skills

Programming & Data	Python, GoLang, C++, Java, Bash, NoSQL
Simulation & Data	Aspen Plus, COMSOL, SolidWorks, MATLAB, JMP, Excel
AI & Automation	PyTorch, Hugging Face, LangChain, OpenCV, Azure, AWS Lambda
Certifications & WIP	FE Chemical, AutoCAD Certified Associate
Interests	Soccer, Formula 1, Cars, Racing Physics, Hiking, FIFA, Smite